

# Indian Standards (BIS) & Regulatory Specification for Electric Immersion Water Heaters

**Target Product:** Electric Immersion Water Heater (Model EWH-1500, 1500W, 230V AC, 50Hz) | **Zyntrix Regulatory Ref:** ZYN-BIS-EWH-2026

|                                                              |                                                                |                                               |                                                    |
|--------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------|
| <b>Product Category:</b><br>Electrical & Domestic Appliances | <b>Compulsory Scheme:</b><br>Scheme-I (ISI Mark Certification) | <b>Primary Standard:</b><br>IS 302-2-201:2008 | <b>Statutory Mandate:</b><br>DPIIT QCO S.O. 189(E) |
|--------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------|

## 1. Knowledge Base Analysis & Regulatory Distinction

During verification of the repository knowledge base against the sample test report (Electric\_Immersion\_Water\_Heater\_Lab\_Report.pdf), a crucial technical distinction was identified between general liquid heaters and electric immersion water heaters:

**Current Knowledge Base State:** Zyntrix's verified dataset (data/bis/verified/) contains **IS 302-2-15:2009** (*Appliances for Heating Liquids*). While IS 302-2-15 covers kettles, coffee makers, and boiling urns, it does not legally or technically govern portable immersion rods.

**The Correct Governing Standard:** Portable immersion rods are strictly regulated under **IS 302 (Part 2/Sec 201) : 2008** (*Particular Requirements for Electric Immersion Water Heaters*), read in conjunction with general safety standard **IS 302-1:2008**.

**Supersession History:** IS 302 (Part 2/Sec 201):2008 formally superseded the older standard **IS 368:1992**. Immersion heaters certified today must conform to IS 302-2-201:2008.

**Mandatory Quality Control Order (QCO):** Immersion water heaters are governed by the Central Government's **Electrical Appliances (Quality Control) Order, 2003 (S.O. 189(E))** issued by DPIIT. It is statutory and compulsory — no immersion rod can be produced, imported, or sold without the BIS ISI Mark.

## 2. Official Indian Standards (BIS) Master Matrix

The following Indian Standards constitute the complete mandatory compliance package for electric immersion water heaters:

| Standard Designation           | Title / Description                                                                                                                     | Role & Applicability                                                                                                                                            | Regulatory Status                                               |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| IS 302 (Part 2/Sec 201) : 2008 | Safety of Household and Similar Electrical Appliances — Part 2: Particular Requirements — Section 201: Electric Immersion Water Heaters | <b>Primary Product Standard:</b> Dictates submersion dielectric limits, liquid depth markings, dry-boil safety, and handle insulation.                          | <b>MANDATORY</b><br>Scheme-I (ISI Mark)<br>QCO S.O. 189(E)      |
| IS 302-1 : 2008 / 2024         | Safety of Household and Similar Electrical Appliances — Part 1: General Requirements                                                    | <b>General Safety Standard:</b> Read in conjunction with Part 2/Sec 201. Covers rated power tolerances, insulation resistance, electric strength, and earthing. | <b>MANDATORY</b><br>Base standard for all electrical appliances |

|                            |                                                                                           |                                                                                                                                                           |                                                                    |
|----------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| IS 1293 : 2019             | Plugs and Socket-Outlets for Household and Similar Purposes of Rated Voltage up to 250V   | <b>Subcomponent Standard:</b> The 3-pin 6 A or 16 A molded plug fitted on the heater power cord must carry independent ISI marking.                       | <b>MANDATORY</b><br>Plugs & Sockets QCO<br>Independent ISI License |
| IS 694 : 2010              | Polyvinyl Chloride (PVC) Insulated Cables for Working Voltages up to and Including 1100 V | <b>Subcomponent Standard:</b> Flexible 3-core power supply cord (minimum 0.75 mm <sup>2</sup> / 1.0 mm <sup>2</sup> copper conductor) must be ISI marked. | <b>MANDATORY</b><br>Cables QCO<br>Independent ISI License          |
| PM/IS 302 (Part 2/Sec 201) | BIS Product Manual for Electric Immersion Water Heaters                                   | <b>Audit &amp; Testing Manual:</b> Specifies Scheme of Inspection & Testing (SIT), factory routine checks, acceptance testing lots, and sample sizes.     | <b>STATUTORY BIS</b><br>Audit & Licensing Manual                   |
| IS 4905 : 2015             | Random Sampling and Randomization Procedures                                              | <b>Method Standard:</b> Statistical sampling framework for production batch lot testing and third-party NABL lab surveillance.                            | Sampling Guideline                                                 |

3. Clause-by-Clause Compliance Mapping (Lab Report Evaluation)

Cross-mapping of test parameters from Electric\_Immersion\_Water\_Heater\_Lab\_Report.pdf (Model EWH-1500, 1500W, 230V AC) against statutory Indian Standard clauses, test conditions, and compliance thresholds:

| Lab Parameter          | Lab Result                     | Governing IS Clause                           | Statutory Permissible Limit / Criterion                                                                                      | Verdict                            |
|------------------------|--------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| Rated Power Test       | 1492 W (@ 230V AC)             | IS 302-1 Clause 10 (Rating & Input)           | Permissible deviation for > 200W heating appliances is <b>+5% to -10%</b> (1350 W to 1575 W).                                | <b>PASS</b><br>(-0.53% dev)        |
| Input Voltage          | 230 V AC, 50 Hz                | IS 302-1 Clause 6                             | Standard single-phase rated AC supply voltage for Indian grid (230 V ± 10%).                                                 | <b>PASS</b>                        |
| Insulation Resistance  | 25 MΩ                          | IS 302-1 Clause 13 & 16 (Moisture & Electric) | Minimum acceptable insulation resistance is ≥ <b>2.0 MΩ</b> (cold) and ≥ <b>5.0 MΩ</b> after conditioning.                   | <b>PASS</b><br>(5x margin)         |
| Electric Strength Test | No Breakdown (@ 1500 V AC, 1m) | IS 302-1 Clause 13 & 16 & IS 302-2-201        | No flashover or dielectric breakdown under <b>1500 V AC sinusoidal</b> applied for 60 seconds.                               | <b>PASS</b><br>(Dielectric intact) |
| Leakage Current Test   | 0.32 mA                        | IS 302-1 Clause 13                            | Maximum permissible leakage current for Class I portable appliances is ≤ <b>0.75 mA</b> .                                    | <b>PASS</b><br>(Well within limit) |
| Earthing Continuity    | 0.08 Ω                         | IS 302-1 Clause 27 (Provision for Earthing)   | Resistance between earth pin and accessible metal parts shall not exceed <b>0.10 Ω</b> (at 10 A or 25 A test current).       | <b>PASS</b><br>(< 0.10 Ω limit)    |
| Temperature-Rise       | Within limit (Handle safe)     | IS 302-1 Clause 11 (Heating)                  | Max temperature rise for non-metallic grip/handle is ≤ <b>60 K</b> . Terminal connections must not overheat.                 | <b>PASS</b>                        |
| Mechanical Strength    | No damage (After drop/impact)  | IS 302-1 Clause 21 (Mechanical)               | Withstand 3 impacts of 0.5 J using spring hammer; drop test from 1.0 m height without exposing live parts.                   | <b>PASS</b>                        |
| Marking & Labeling     | Compliant                      | IS 302-1 Clause 7 & IS 302-2-201 Clause 7     | Mandatory indelible markings: V, W, Model, Maker, <b>Min &amp; Max Immersion Level lines</b> , and ISI Standard Mark + CM/L. | <b>PASS</b>                        |

|                               |                            |                                              |                                                                                                                        |      |
|-------------------------------|----------------------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------|
| Submersion Ingress & Safety   | Compliant (Immersion Zone) | IS 302-2-201 Clause 15 (Moisture Resistance) | Water immersion up to maximum water line for 24 hours followed by insulation resistance and dielectric strength check. | PASS |
| Abnormal Operation (Dry-Boil) | No Hazard (Thermal Cutout) | IS 302-2-201 Clause 19 (Abnormal Operation)  | Energizing heater out of water for prescribed duration without burst, molten metal emission, or flame propagation.     | PASS |

4. Factory Quality Control & Scheme of Inspection and Testing (SIT)

Under BIS Product Manual PM/IS 302 (Part 2/Sec 201), manufacturers must enforce a three-tier testing regime in their factory quality control laboratory to maintain ISI Mark certification:

| Testing Category | Sampling Frequency                              | Mandatory Tests Included                                                                                                                       | Acceptance Criteria                                                    |
|------------------|-------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Routine Tests    | 100% of Production (Every manufactured unit)    | 1. Earth continuity test (0.1 Ω limit)<br>2. High-voltage electric strength (1500V, 1 sec flash)<br>3. Power consumption test at rated voltage | Zero tolerance. Defective units rejected immediately before packaging. |
| Acceptance Tests | Per Lot / Batch (Random sample per IS 4905)     | 1. Insulation resistance (cold & warm)<br>2. Leakage current test (≤ 0.75 mA)<br>3. Physical dimension and marking verification                | Lot accepted if samples pass; batch quarantined if defect rate > AQL.  |
| Type Tests       | Qualification & Annual (Full independent audit) | All clauses (Clause 7 to Clause 32), including abnormal operation (dry boil), mechanical drop, heating, and endurance.                         | Must be performed in BIS-approved or NABL accredited laboratory.       |

5. Knowledge Base Upgrade Specification for Zyntrix Platform

To enable the Zyntrix platform to automatically and accurately evaluate Electric Immersion Water Heaters with 100% precision, the following verified knowledge base artifacts have been drafted for ingestion:

- 1. **Verified Standard Folder:** Create data/bis/verified/IS\_302\_2\_201\_2008/ containing metadata.json, regulatory/qco\_order\_electrical.json, and product\_manual/pm\_is302\_2\_201.json.
- 2. **Applicability Rule Addition:** Add rule APP-ELECTRICAL-002 specifically targeting keywords ['immersion heater', 'immersion rod', 'water heating rod'] mapping to IS 302-2-201:2008.
- 3. **Auxiliary Subcomponent Linking:** Link prerequisite verification for 3-pin plug (IS 1293) and flexible cord (IS 694) in the Digital Compliance Passport.
- 4. **Lab Evidence Clause Recalculator:** Map uploaded lab report parameters directly to clauses 7, 10, 11, 13, 15, 16, 19, 21, and 27 for instant zero-drift verdict determination.

**Statutory Notice & Authority:** This technical specification is compiled in accordance with the Bureau of Indian Standards Act, 2016 and the Electrical Appliances (Quality Control) Order, 2003 (S.O. 189(E)). Generated by Zyntrix AI Compliance Compiler (SIH Problem 26107). Verified against official BIS Gazette notifications.